FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 05:09:47 to 17:00:33 UTC | 200 sps | logger 99977

Folder: 2026-10-30_FW0860_deck_PPS_200sps
Pulses analyzed: ADC1 42641, ADC2 42640, ADC3 42641, ADC4 42641

Timing (negative offset = recorded edge later than reference; RMS resid = raw/after per-pulse shift)

ADC	chan	n	offset med (std) ms	theory ms	obs-theory	best-fit shift med (std) ms	RMS resid %
ADC1	HDF	42641	-0.289 (0.073)	+0.000	-0.289	-0.261 (0.087)	0.89/ 0.52
ADC2	EHZ	42640	-8.217 (1.441)	-5.994	-2.223	-2.236 (1.432)	4.88/ 0.38
ADC3	EHN	42641	-8.037 (1.439)	-5.994	-2.044	-2.043 (1.430)	4.59/ 0.38
ADC4	EHE	42641	-7.886 (1.439)	-5.994	-1.892	-1.883 (1.431)	4.37/ 0.39

ADC clocks relative to ADC1 (from the sawtooth of the ADC2-4 offsets)

ADC	chan	clock - ADC1, ppm: median	5-95% over run	ADC clock vs TCXO, ppm
ADC2	EHZ	-319 (alias limit ±2500)	-491 ... -267	-33
ADC3	EHN	+494 (alias limit ±2500)	+390 ... +538	+780
ADC4	EHE	+669 (alias limit ±2500)	+466 ... +742	+955

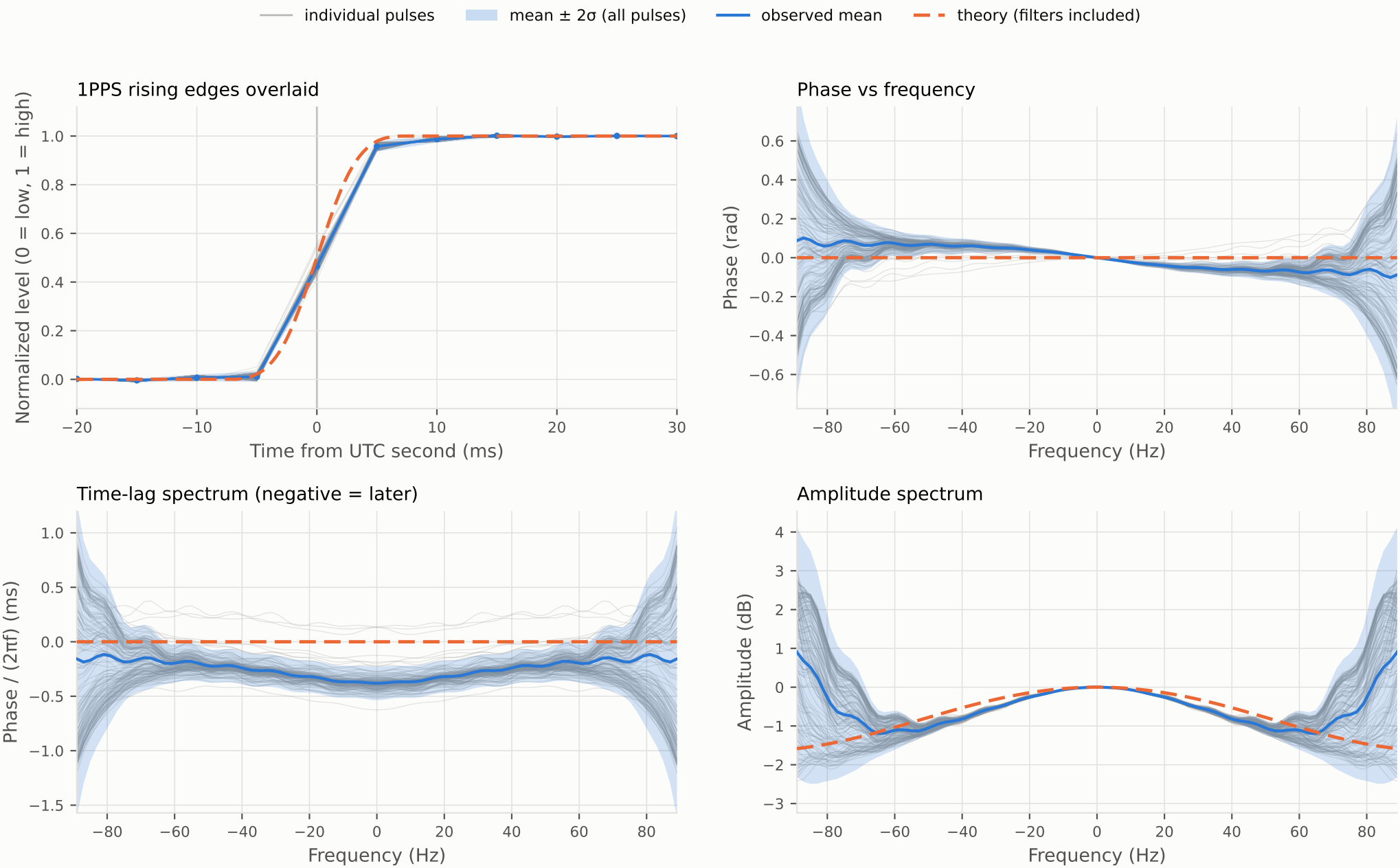
ADC1 clock vs TCXO (log adc_ppm median): +286 ppm. ADC1 offset trend: +0.9 us/hour.

How to read this report

- Theory = the signal that should have been recorded: ideal 1PPS step through the analog filter (ADC2-4 only) and the ADC sinc3 filter, sampled on the UTC grid, then passed through the same analysis as the data (levels, padding, aliasing).
- sinc3 is zero-phase: firmware time-tags each sample at the center of the sinc3 window.
- ADC2-4 analog model: ladder, corner 48.25 Hz per stage, stage-2 impedance 10 x stage 1.
- Edge plots: time axis is ((t + 0.5) mod 1) - 0.5 seconds from the UTC second. Spectra: DFT of each pulse relative to an ideal step (periodic square-wave padding), so only odd harmonics of 1 Hz exist; negative frequencies mirror positive ones; phase is unwrapped outward from low frequency. Envelope = mean ± 2 std over ALL pulses (<=150 individual drawn).
- Offset = phase slope through the origin below 40 Hz. Best-fit shift = time shift of the theoretical edge that minimizes squared error against each observed edge in the time domain.
- ADC2-4 are read at ADC1's DRDY while converting on their own clocks, so each sample is 0..T older than ADC1's (T = sample period): expected mean extra lag T/2 and offset scatter T/sqrt(12) = 1.443 ms (observed std: ADC2 1.441, ADC3 1.439, ADC4 1.439 ms).
- Clock beat: the offset sawtooth advances (clock difference x 1 s) per pulse, wrapped to +/- T/2. Sign: a faster ADC clock gives older data, so clock difference (ppm) = -(offset increment, ms/s) x 1000. Valid within the alias limit.

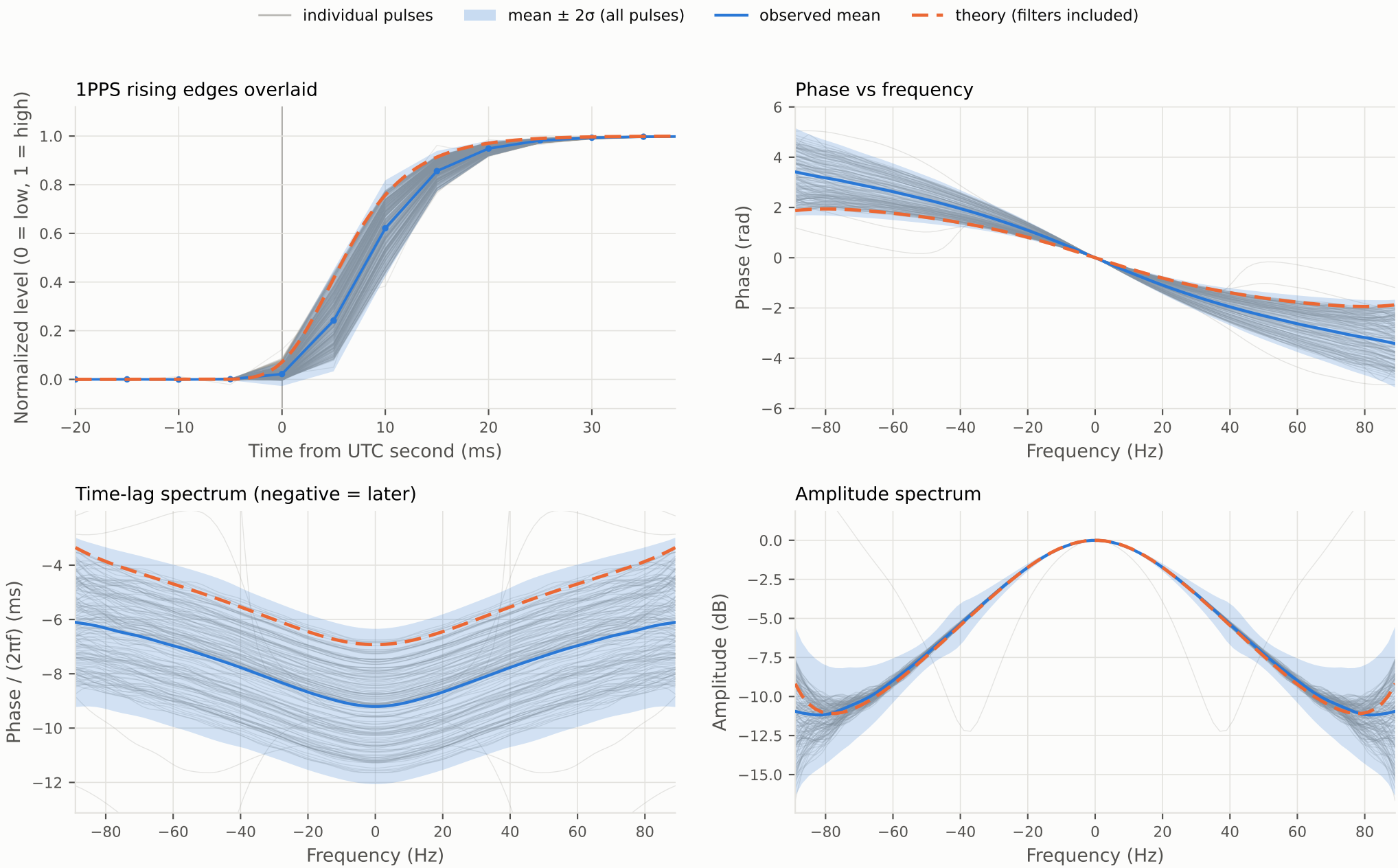
ADC1 (HDF): no analog filter + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 05:09:47 to 17:00:33 UTC | 200 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -0.289 ms (std 0.073)   theory +0.000 ms   observed - theory -0.289 ms
Per-pulse best-fit shift of theoretical edge: median -0.261 mean -0.257 std 0.087 2.5-97.5%: [-0.394, -0.092] ms
Edge residual RMS (% of step): 0.89 raw, 0.52 after per-pulse shift.   Mean spectrum vs theory, RMS: phase 0.061 rad, amplitude 0.73 dB



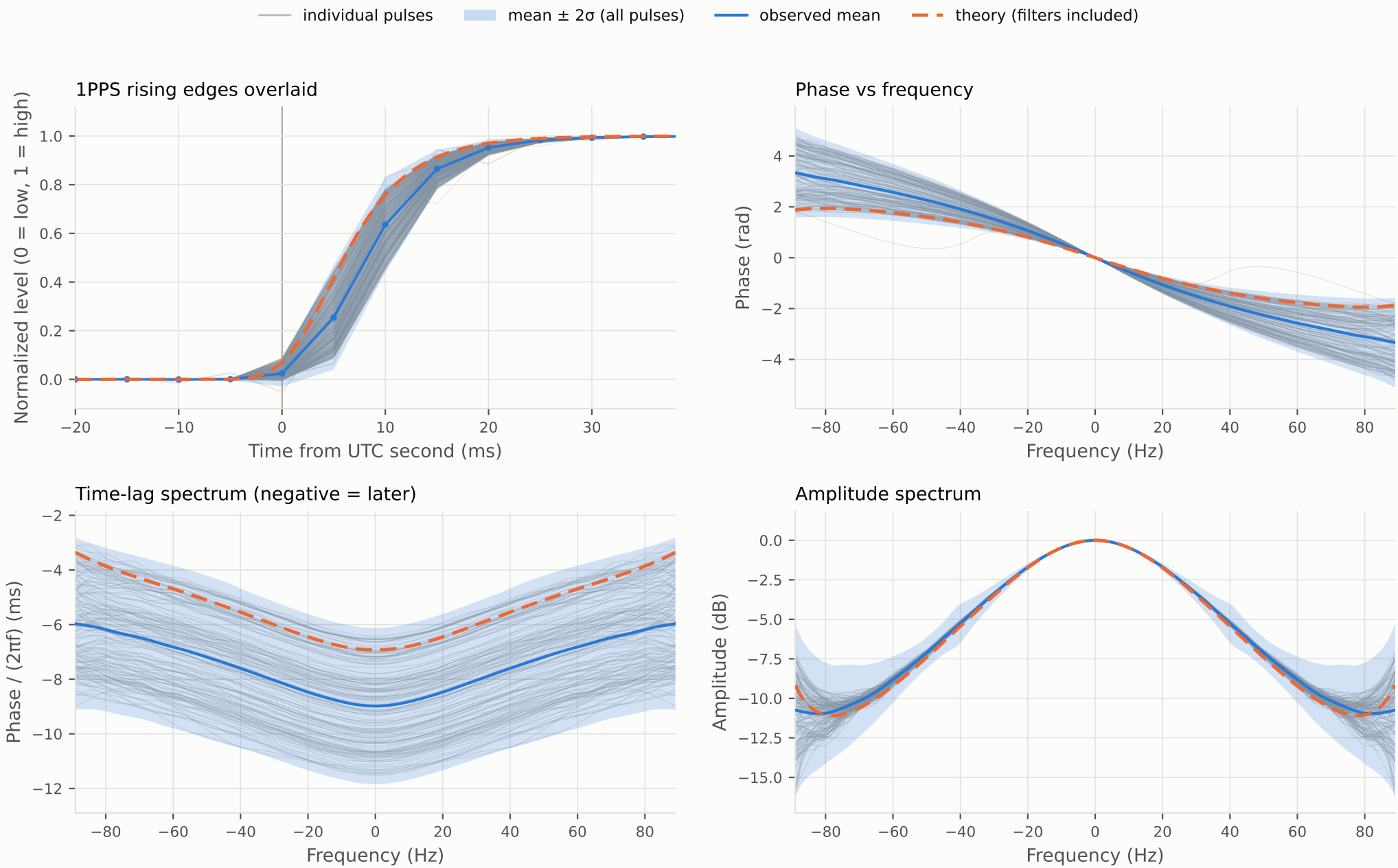
ADC2 (EHZ): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 05:09:47 to 17:00:33 UTC | 200 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -8.217 ms (std 1.441) theory -5.994 ms observed - theory -2.223 ms
Per-pulse best-fit shift of theoretical edge: median -2.236 mean -2.242 std 1.432 2.5-97.5%: [-4.603, +0.111] ms
Edge residual RMS (% of step): 4.88 raw, 0.38 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.783 rad, amplitude 0.36 dB



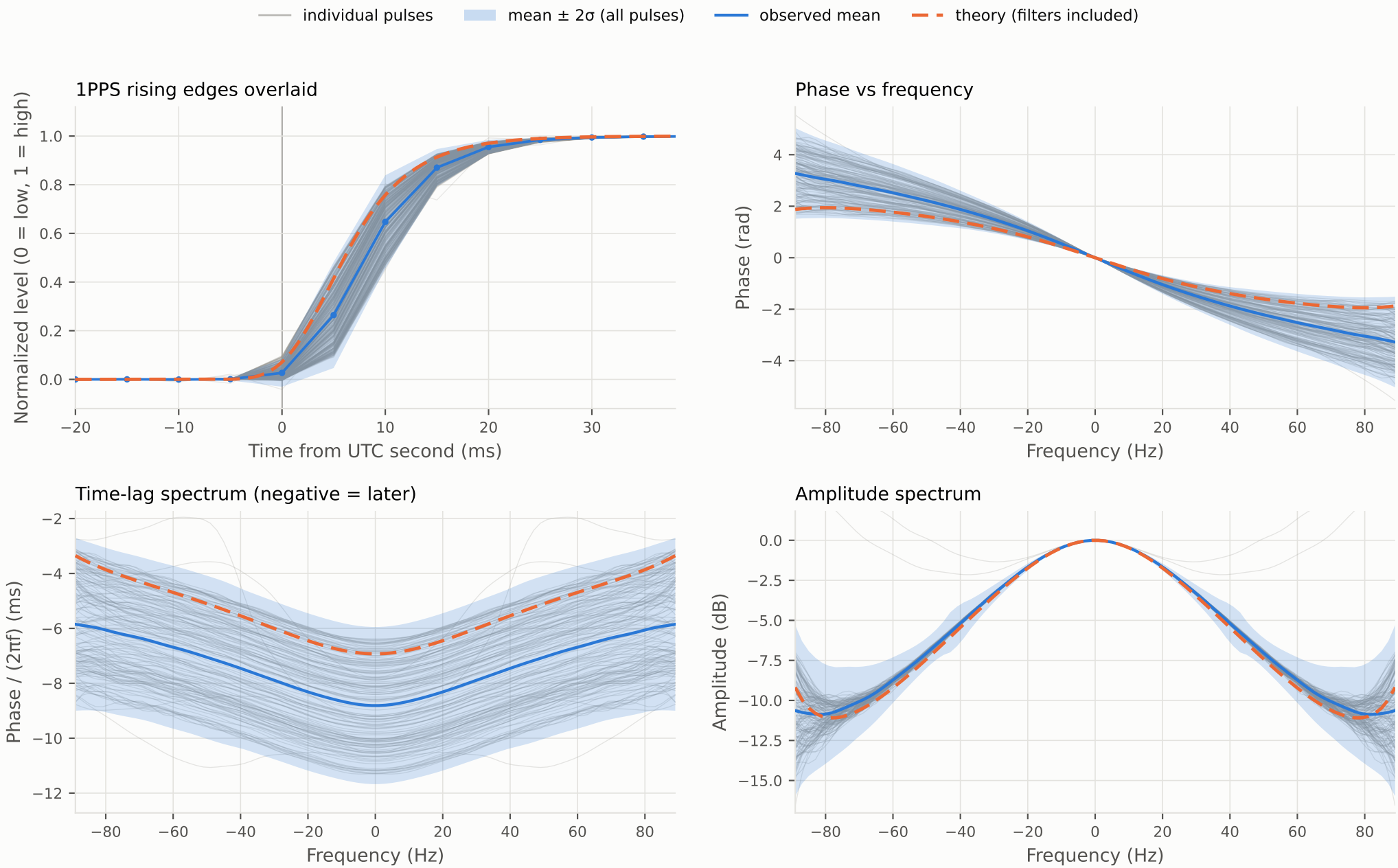
ADC3 (EHN): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 05:09:47 to 17:00:33 UTC | 200 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -8.037 ms (std 1.439) theory -5.994 ms observed - theory -2.044 ms
Per-pulse best-fit shift of theoretical edge: median -2.043 mean -2.042 std 1.430 2.5-97.5%: [-4.396, +0.315] ms
Edge residual RMS (% of step): 4.59 raw, 0.38 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.735 rad, amplitude 0.37 dB



ADC4 (EHE): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 05:09:47 to 17:00:33 UTC | 200 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -7.886 ms (std 1.439) theory -5.994 ms observed - theory -1.892 ms
Per-pulse best-fit shift of theoretical edge: median -1.883 mean -1.882 std 1.431 2.5-97.5%: [-4.240, +0.472] ms
Edge residual RMS (% of step): 4.37 raw, 0.39 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.694 rad, amplitude 0.39 dB



Time offset and ADC clock beat vs time

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 05:09:47 to 17:00:33 UTC | 200 sps | logger 99977

Offsets from the phase slope below 40 Hz (negative = recorded edge later); dashed gray = theory.

